

Rejection Sampling: Cauchy(5) Proposal Derivation

leta199

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1 Introduction

For this next proposal, we will consider shifting a Cauchy distribution to be centred at $x = 5$. We will also find the necessary scaling factor M for our Cauchy to have the proposal envelope the target at all points.

$$\begin{aligned}\frac{f(x)}{g(x)} &= \frac{\left(\frac{xe^{-x}}{6e^{-5}}\right)}{\frac{1}{\pi(1+(x-5)^2)}} \\ &= \frac{\pi}{6e^{-5}}(xe^{-x})(1+(x-5)^2) \\ &= \frac{\pi}{6e^{-5}}e^{-x}x(1+(x^2-10x+25)) \\ &= \frac{\pi}{6e^{-5}}e^{-x}x(x^2-10x+26) \\ &= \frac{\pi}{6e^{-5}}e^{-x}(x^3-10x^2+26x)\end{aligned}$$

Derivative to find max value of ratio:

Let:

$$\begin{aligned}u &= \frac{\pi}{6e^{-5}}e^{-x} & v &= x^3 - 10x^2 + 26x \\ du &= -\frac{\pi}{6e^{-5}}e^{-x} & dv &= 3x^2 - 20x + 26\end{aligned}$$

Using the product rule:

$$\begin{aligned}\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \frac{\pi}{6e^{-5}}e^{-x}(3x^2 - 20x + 26) - \frac{\pi}{6e^{-5}}e^{-x}(x^3 - 10x^2 + 26x) \\ &= \frac{\pi}{6e^{-5}}e^{-x}(-x^3 + 3x^2 + 10x^2 - 20x - 26x + 26) \\ &= \frac{\pi}{6e^{-5}}e^{-x}(-x^3 + 13x^2 - 46x + 26) \\ &= -\frac{\pi}{6e^{-5}}e^{-x}(x^3 - 13x^2 + 46x - 26)\end{aligned}$$

To find the critical points, set the derivative to zero:

$$\frac{\pi}{6e^{-5}}e^{-x} = 0 \quad (\text{No Solution}) \quad \text{or} \quad x^3 - 13x^2 + 46x - 26 = 0$$

We can then find the solution to the cubic via optimisation in R.